

Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



ck800wa

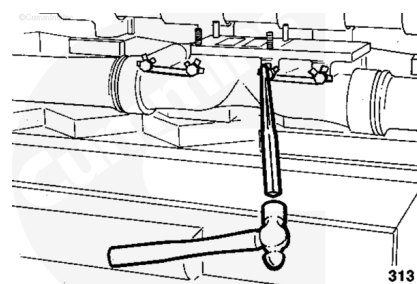
- Disconnect the batteries or air supply line to the air starter to prevent accidental engine starting. Refer to Procedure 013-009 in Section 13 (</qs3/pubsys2/xml/en/procedures/99/99-013-009.html>)
- Drain the cooling system. Refer to Procedure 008-018 in Section 8 (</qs3/pubsys2/xml/en/procedures/20/20-008-018-tr.html>)
- Remove the water bypass clamp. Loosen both hose clamps. Remove the bypass tube, hose, and water connection.
- Remove the turbocharger heat shield, if equipped. Refer to Procedure 010-076 in Section 10 (</qs3/pubsys2/xml/en/procedures/20/20-010-076-tr.html>)
- Remove the turbocharger. Refer to Procedure 010-033 in Section 10 (</qs3/pubsys2/xml/en/procedures/20/20-010-033-tr.html>)

Remove

All Applications Except Marine

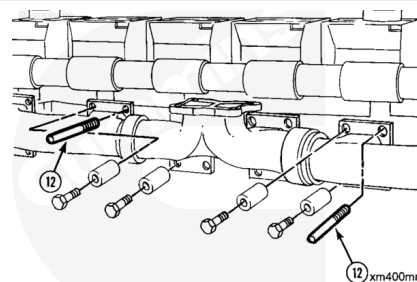
Use a drift and hammer to bend the lock plate tabs away from the capscrew heads, if equipped.

Note : Some engines do not use this type of lock plate. Some engines use a lock plate that snaps over the capscrew head. Some engines use locking capscrews that do not require lockplates.



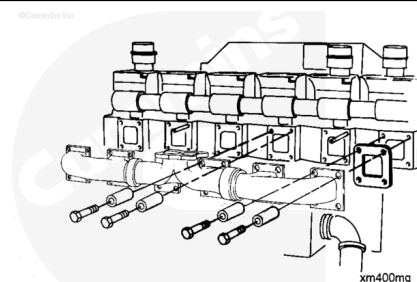
Remove two capscrews as shown.

Install two 11.112 mm x 127 mm [0.4375 in x 5 in] guide studs (12), as shown.



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



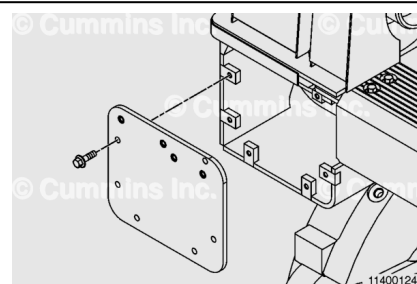
Remove the remaining capscrews.

Remove the exhaust manifold and the gaskets.

Discard the gaskets.

Marine Applications

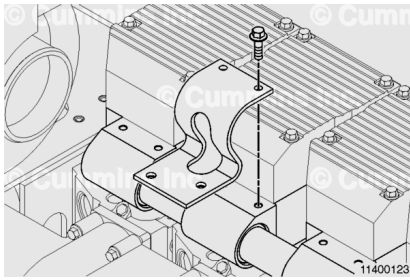
Remove the six mounting capscrews and the turbocharger bracket cover plate.



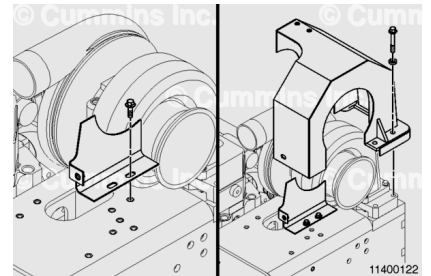
Remove the exhaust manifold heat shield. Refer to Procedure 011-015 in Section 11.

(</qs3/pubsys2/xml/en/procedures/20/20-011-015-tr.html>)

Remove the exhaust manifold heat shield bracket and capscrews from the rocker box.

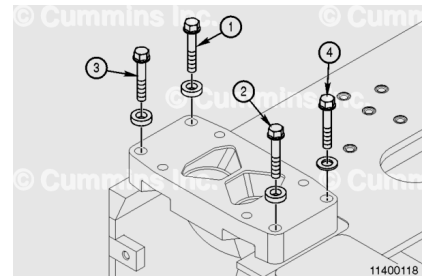


Remove the turbocharger heat shield support bracket.



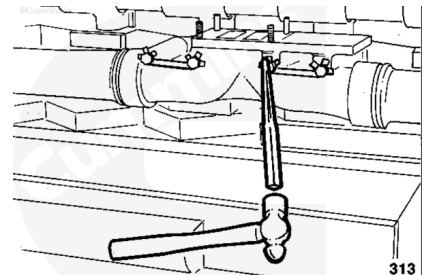
Remove capscrews 1, 2, and 3, and spacers.

Remove capscrew 4 and washer.



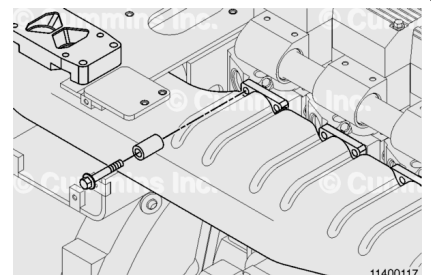
Use a drift and hammer to bend the lock plate tabs away from the capscrew heads, if equipped.

Note : Some engines do not use this type of lock plate. Some engines use a lock plate that snaps over the capscrew head. Some engines use locking capscrews that do not require lockplates.

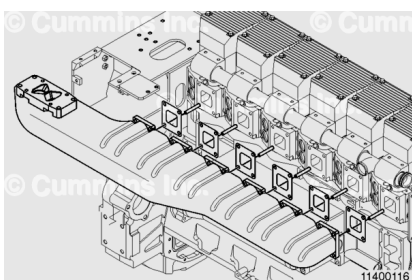


Remove the lower capscrews and install alignment rods.

Remove the 12 upper exhaust manifold mounting capscrews and spacers.



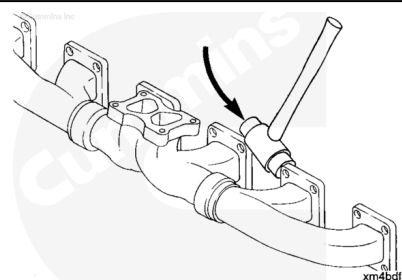
Remove the exhaust manifold, with gaskets, from the cylinder heads.



Disassemble

All Applications Except Marine

Use a mallet to remove the end sections from the exhaust manifold center section.



Clean and Inspect for Reuse

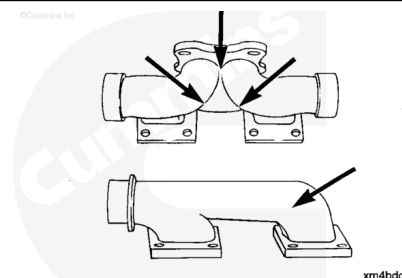
All Applications

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.



Use solvent or steam to clean the exhaust manifold.

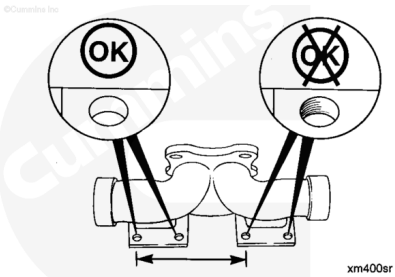
Use a wire brush to remove any scale on the exhaust manifold sealing joint inside and outside diameters.

Inspect the exhaust manifold for cracks.

Inspect the capscrew holes in the center section for damage.

The exhaust manifold is subjected to high temperatures that cause the manifold to shrink. Capscrew threads visible on the side of the capscrew holes is an indication the exhaust manifold has shrunk.

Measure the center-to-center distance between the same position capscrew holes of the two flanges on the center section and both end sections to determine if the manifold is too short.



xm400sr

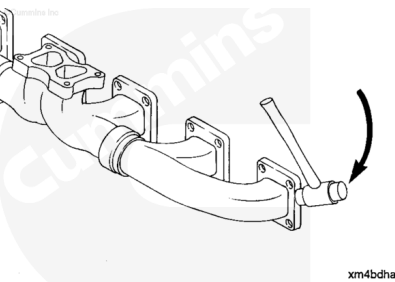
Exhaust Manifold Capscrew Holes

mm		in
191.5	MIN	7.54
193.5	MAX	7.62

Assemble

All Applications Except Marine

Use a mallet and drive the end section into the center section.



xm4bdha

Install

All Applications Except Marine

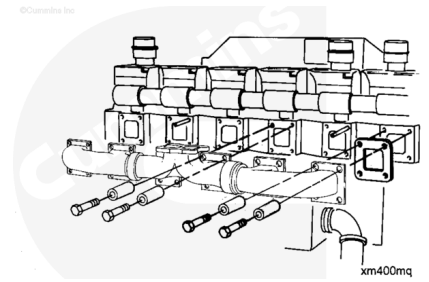
⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this

component or assembly.

⚠ CAUTION ⚠

Do not use gasket cement when installing the exhaust manifold gasket. Use of gasket cement will cause the gasket to fail.



Install the guide studs used during removal.

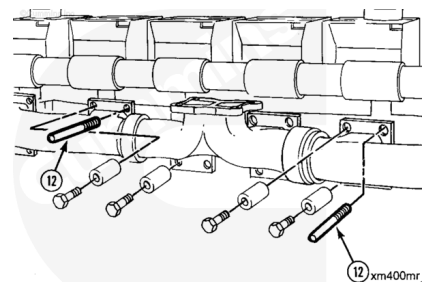
Use contact adhesive and attach the gasket on the cylinder head exhaust ports.

Install the exhaust manifold.

Apply an anti-seize compound on the capscrew threads.

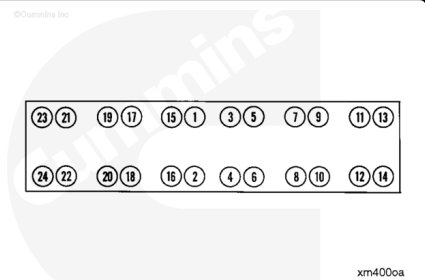
Install the capscrews.

Remove the two guide studs (12) and install the remaining capscrews.



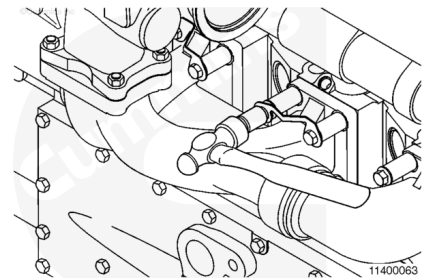
Tighten the capscrews in the sequence shown.

Torque Value: 61 n•m [45 ft-lb]



⚠ CAUTION ⚠

When installing the lock plate that snaps over the capscrew head, install the lock plate approximately one-half of the capscrew head height. Installing the lock plate until it bottoms will cause it to deform.



Note : Some engines do not use this type of lock plate. Some engines use a lock plate that snaps over the capscrew head. Some engines use locking capscrews that do not require lockplates.

Align the lock plate with the top of both capscrews.

Place the socket that was used to install the capscrew onto one of the capscrews and drive the lock plate down halfway the capscrew flange.

Repeat the process on the remaining capscrews and lock plates.

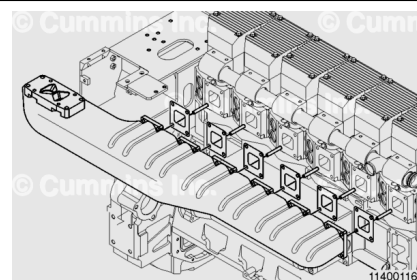
Marine Applications

Insert the alignment rods for the lower exhaust manifold mounting holes into the cylinder heads.

Install the exhaust manifold, with gaskets, to the cylinder heads on the alignment rods.

Alignment rods provide support during installation.

The exhaust manifold turbocharger flange **must** sit on the turbocharger support bracket with all four mounting holes in their approximate positions.

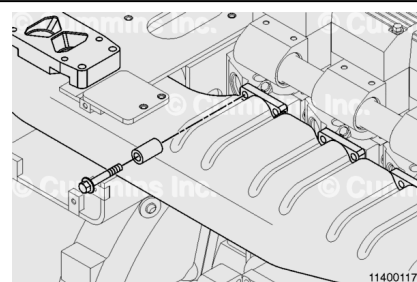


Assemble one manifold mounting capscrew to one manifold mounting spacer. Twenty-four of these assemblies are needed.

Apply high temperature copper anti-seize compound to the threads of the 24 exhaust manifold mounting capscrews.

Loosely install the 12 upper exhaust manifold mounting capscrews with spacers. Engage at least 5 threads.

Remove the alignment rods from the lower holes and loosely install the 12 lower manifold mounting capscrews with spacers. Engage at least 5 threads.



Apply anti-seize compound to the threads of capscrews 1, 2, 3, and 4.

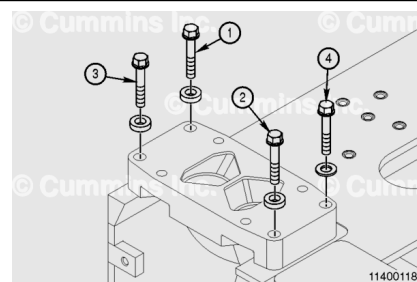
Manually align the holes in the manifold to the holes in the bracket.

Loosely install capscrews 1, 2, or 3, with spacer, into the holes. Engage at least 5 threads.

Install capscrew 4 with a washer.

Tighten all four capscrews finger-tight.

Use caution when tightening capscrews 1 and 2. If the capscrews meet resistance before the capscrew head and washer contact the bracket, remove them before tightening

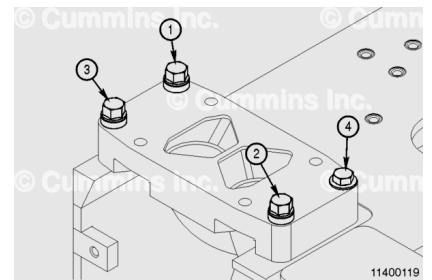


any farther. If this occurs, chase the threads with a 7/16-in-14 UNC 2B tap to clean the threads.

Tighten all four capscrews in the sequence shown.

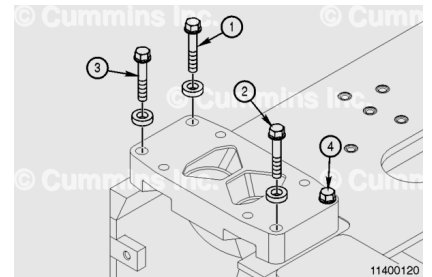
Torque Value: 45 n•m [33 ft-lb]

Capscrew 4 can **not** be installed, tightened, or removed after the turbocharger is installed.



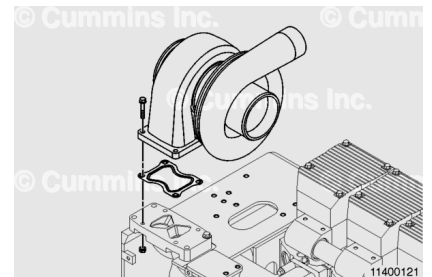
Remove capscrews and spacers 1, 2, and 3. Leave capscrew 4 installed and tightened.

Place all necessary turbocharger oil and coolant supply and drain hoses through the holes in the bracket. These will be mounted to the turbocharger and engine later, but **must** be in place before the turbocharger so there is room to maneuver the hoses into place.



Install the turbocharger. Refer to Procedure 010-033 in Section 10. (</qs3/pubsys2/xml/en/procedures/20/20-010-033-tr.html>)

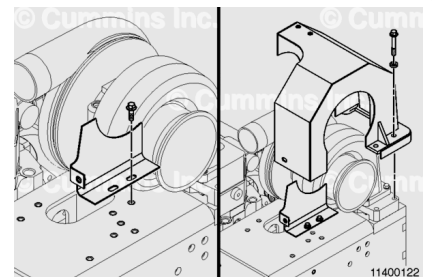
Do **not** install the exhaust system to the turbocharger at this time.



Install the turbocharger heat shield support bracket.

Torque Value: 35 n•m [26 ft-lb]

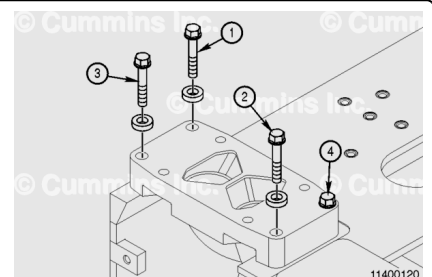
Install the turbocharger heat shield. Refer to Procedure 010-076 in Section 10. (</qs3/pubsys2/xml/en/procedures/20/20-010-076-tr.html>)



Make sure holes 1, 2, and 3 are still aligned, otherwise they **must** be manually realigned.

Capscrews 1, 2, and 3 **must** still have an adequate amount of anti-seize compound on the threads. Apply anti-seize compound, if needed.

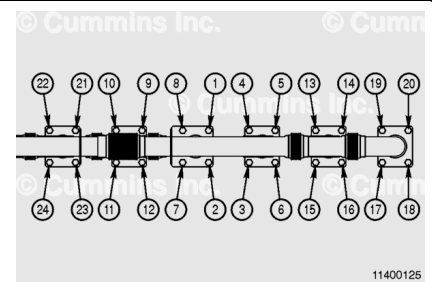
Install capscrews and spacers 1, 2, and 3.



Torque Value: 45 n•m [33 ft-lb]

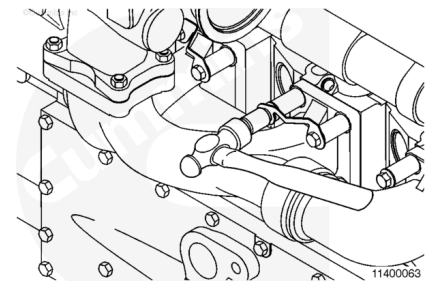
Tighten the exhaust manifold mounting capscrews in the sequence shown.

Torque Value: 45 n•m [33 ft-lb]



⚠ CAUTION ⚠

When installing the lock plate that snaps over the capscrew head, install the lock plate approximately one-half of the capscrew head height. Installing the lock plate until it bottoms will cause the lockplate to deform.



Note : Some engines do not use this type of lock plate. Some engines use a lock plate that snaps over the capscrew head. Some engines use locking capscrews that do not require lock plates.

Align the lock plate with the top of both capscrews.

Place the socket that was used to install the capscrew onto one of the capscrews and drive the lock plate halfway down the capscrew flange.

Repeat the process on the remaining capscrews and lock plates.

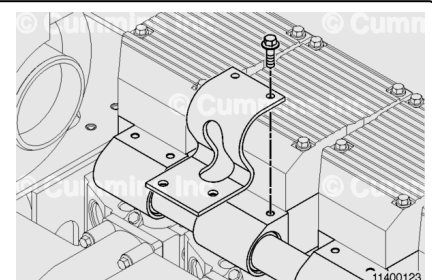
Apply anti-seize compound to the threads of the two exhaust manifold heat shield bracket capscrews.

Install the exhaust manifold heat shield bracket and capscrews to the rocker box.

Leave the capscrews loose until the heat shield is installed.

Install the exhaust manifold heat shield. Refer to Procedure 011-015 in Section 11.

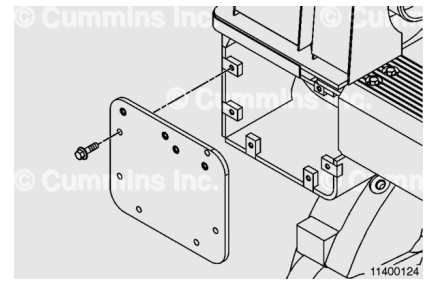
(qs3/pubsys2/xml/en/procedures/20/20-011-015-tr.html)



Apply anti-seize compound to the threads of the turbocharger bracket cover plate capscrews.

Install the turbocharger bracket cover plate, six capscrews and tighten.

Torque Value: 35 n·m [26 ft·lb]



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



- Install the turbocharger. Refer to Procedure 010-033 in Section 10 (</qs3/pubsys2/xml/en/procedures/20/20-010-033-tr.html>) if **not** completed.
- Install the turbocharger heat shield, if removed. Refer to Procedure 010-076 in Section 10 (</qs3/pubsys2/xml/en/procedures/20/20-010-076-tr.html>) if **not** completed.
- Install the bypass tube, hose, and water connection. Install the bypass tube clamp. Tighten both clamps.
- Fill the cooling system. Refer to Procedure 008-018 in Section 8 (</qs3/pubsys2/xml/en/procedures/20/20-008-018-tr.html>)
- Connect the batteries or air supply line to the air starter. Refer to Procedure 013-009 in Section 13 (</qs3/pubsys2/xml/en/procedures/99/99-013-009.html>)
- Operate the engine and check for leaks.

